

RPG FRAMEWORK

CUSTOM STATS IN-GAME EDITOR

Complete UI design book

A Minecraft-native creator workflow for building, validating, testing,
and maintaining data-driven custom stats.

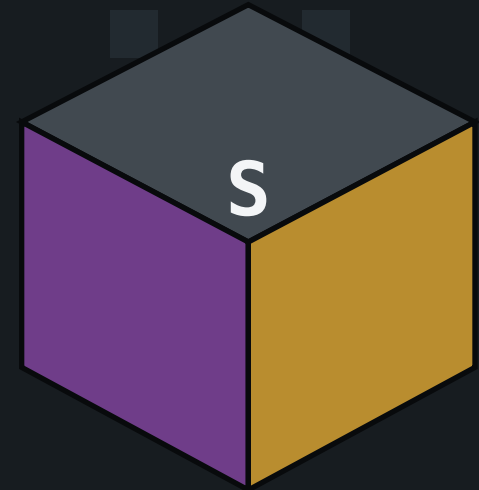
23 SCREEN STATES

8-PIXEL GRID

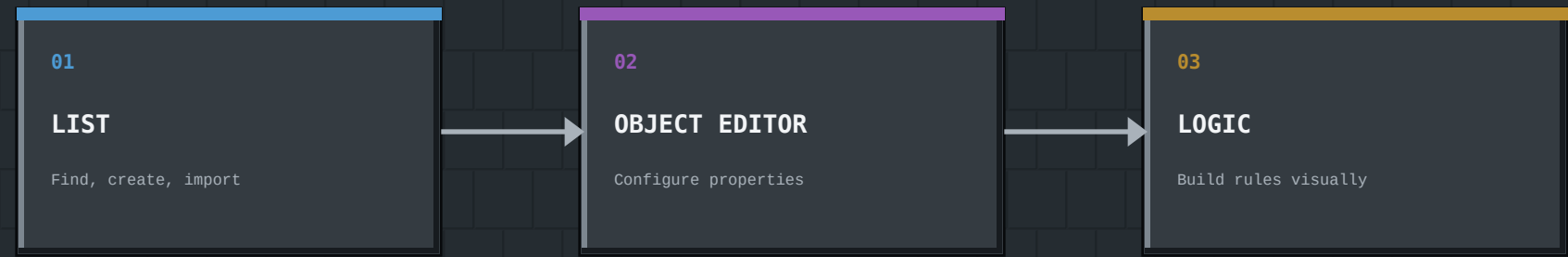
JSON-SAFE

DESIGN DIRECTION

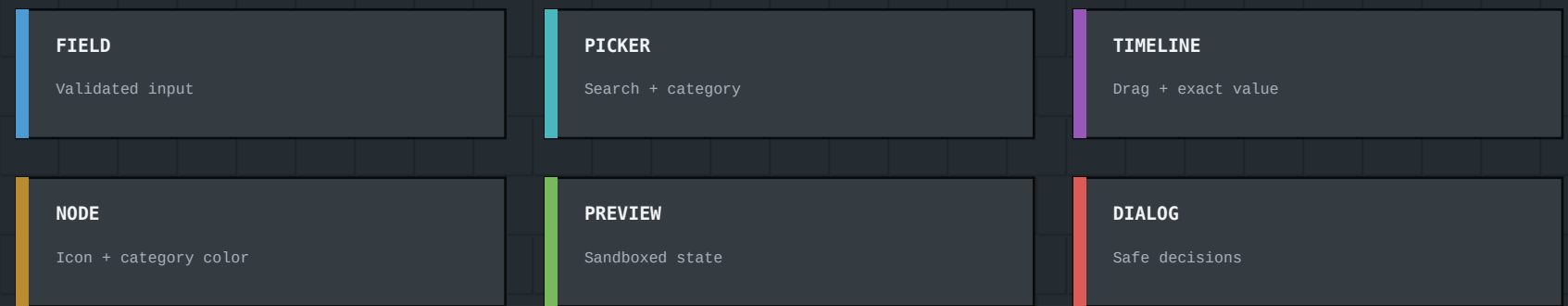
DARK STONE / PIXEL BEVELS / LAYERED COMPLEXITY



EDITOR WORKFLOW



CORE COMPONENTS



8 PX SPACING / 24 PX CONTROLS / SQUARE CORNERS / LIGHT-TOP, DARK-BOTTOM BEVEL

INTENT

Teach one workflow that can later power stats, quests, dialogue, NPCs, and abilities.

KEY STATES

List screen -> object editor -> specialized logic. Basic and advanced modes share the same data.

BUILD NOTES

Keep navigation, save state, validation placement, buttons, fields, and graph colors identical across RPG Framework editors.

CUSTOM STATS

4 LOADED

+ NEW STAT

FILTER

IMPORT

	CORRUPTION mypack:corruption	0 - 100	EDITED	>
	MANA mypack:mana			>
	REPUTATION mypack:reputation			>
	TEMPERATURE mypack:temperature			>

CORRUPTION

OPEN

DUPLICATE

RENAME

COPY ID

OPEN JSON

EXPORT

DELETE

4 RESULTS

INTENT

Scan many stats quickly while keeping invalid and modified definitions obvious.

KEY STATES

Selected row, invalid row, unsaved dot, empty/search states, and right-click actions.

BUILD NOTES

Persist search per session. Rows are 52 px high; entire row opens on left-click. Destructive actions stay at the bottom of the context menu.

CREATE STAT

Start simple. Every value can be changed later.

NAME

Corruption

RESOURCE ID

mypack:corruption

Generated from name

TEMPLATE

Accumulation ▼

MINIMUM

0

MAXIMUM

100

DEFAULT

0

ACCUMULATION

For values that build up over time and become more dangerous.

RANGE	0 - 100
DEFAULT	0
VALUE	STORED
HUD	HIDDEN

PRESET ONLY - NOT A LOCKED TYPE

CANCEL

ENTER

CREATE STAT

INTENT

Create a useful stat in seconds without forcing every property up front.

KEY STATES

Auto-generated ID, template preview, invalid ID, range conflict, and create-ready state.

BUILD NOTES

Name updates ID until the ID is manually edited. Template values remain ordinary editable values after creation.

CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

NO ERRORS

GENERAL

Identity used by files, UI, and references.

DISPLAY NAME

Corruption

RESOURCE ID

mypack:corruption

Changing this ID may break 14 references.

VIEW REFERENCES

DESCRIPTION

Dark influence slowly consuming the player.

Used by cursed zones and forbidden magic.

91 / 256

ICON



SELECT TEXTURE

mypack:textures/gui/stats/corruption.png

COLOR

#9858B8

INTENT

Keep identity and presentation metadata together, with dependency safety around the resource ID.

KEY STATES

Focused field, texture selection, color picker, missing texture, and referenced-ID warning.

BUILD NOTES

ID is editable but never casual: surface reference count before commit and offer migration when supported.

 CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

NO ERRORS

VALUE

Numeric storage, range, and lifecycle behavior.

VALUE TYPE

☒ Stored

☐ Computed

NUMBER TYPE

☒ Integer

☐ Decimal

DEFAULT

MINIMUM

MAXIMUM

0

0

100

☒ Clamp value to range

VALID RANGE

Default 0 is inside 0 - 100

PERSISTENCE

ON DEATH

DUNGEON EXIT

DIMENSION CHANGE

Keep value

Keep value

Keep value

Modes shown here are registered by the active runtime.

INTENT

Define numeric behavior and lifecycle without exposing runtime-only options.

KEY STATES

Stored/computed, integer/decimal, range validation, clamp, and event-specific persistence.

BUILD NOTES

Computed mode replaces Default with an expression field. Unsupported persistence events should not render as disabled noise.

CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

NO ERRORS

CORRUPTION STAGES

+ ADD STAGE

RANGE 0 - 100

0

10

20

50

80

100

CORRUPTED

POSSESSED

LOST

THRESHOLD 20

STAGES

CLEAN

0 - 9

>

TAINTED

10 - 19

>

CORRUPTED

20 - 49

>

POSSESSED

50 - 79

>

LOST

80 - 100

>

INTENT

Make ranges readable at a glance and thresholds fast to manipulate precisely.

KEY STATES

Selected segment, draggable handle, exact value popover, gaps, overlaps, integer/decimal snapping.

BUILD NOTES

Dragging previews changes; release commits one undo action. Keyboard arrows nudge; Shift uses fine movement for decimals.

CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

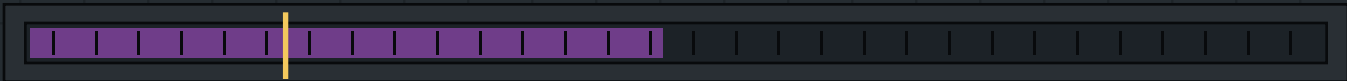
CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

NO ERRORS

STAGES / CORRUPTED



NAME	STAGE ID	FROM	TO
Corrupted	corrupted	20	49

DESCRIPTION

The corruption begins affecting the player's body.

EFFECTS				+ ADD EFFECT	
ATTRIBUTE	Movement speed	-15%	EDIT	X	
	Witch particles	20 ticks	EDIT	X	

EVENTS					
ON ENTER	2 actions	EDIT	ON EXIT	None	EDIT

INTENT

Edit one stage without losing its position in the full stat range.

KEY STATES

Exact start/end, metadata, effects, on-enter/on-exit events, reorder, and delete.

BUILD NOTES

Derived range endpoints remain read-only when adjacent thresholds own them. Effects preserve registered type IDs in serialized JSON.

ADD EFFECT

Search effects...

ALL12

ATTRIBUTES3

STATUS4

VISUAL3

CUSTOM2

✓

Compatible only

ATTRIBUTE MODIFIER
rpg:attribute_modifier

ATTRIBUTE

POTION EFFECT
rpg:potion_effect

STATUS

PERIODIC DAMAGE
rpg:periodic_damage

STATUS

PARTICLES
rpg:particles

VISUAL

RUN ACTION
rpg:run_action

CUSTOM

CANCEL

CONFIGURE

INTENT

Let core modules and addons expose effects in one scalable, searchable catalog.

KEY STATES

Search, categories, compatible-only filter, addon badge, missing dependency, keyboard selection.

BUILD NOTES

Results use friendly labels plus resource IDs. Registry categories determine grouping; unknown types stay visible but cannot be added.

ATTRIBUTE MODIFIER

rpg:attribute_modifier

ATTRIBUTE

Movement Speed ▼

minecraft:generic.movement_speed

OPERATION

Multiply total ▼

VALUE

-0.15

☐ Show raw resource IDs

PREVIEW Movement Speed -15% while stage is active

CANCEL **ADD EFFECT**

INTENT

Use a focused schema-driven form for one registered effect type.

KEY STATES

Friendly/raw ID toggle, invalid resource, conditional fields, edit/add modes, generic fallback form.

BUILD NOTES

The effect registry supplies field schemas and optional custom widgets. Preserve unsupported JSON properties through edits when possible.

 CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

 NO ERRORS

RULES

Automatic ways this stat changes.

+ NEW RULE

01

TRIGGER

EVERY 1200 TICKS

VALID

IF

If Dimension = Overworld

DO

Corruption -1

OPEN >

02

TRIGGER

ON ENTITY KILLED

VALID

IF

If Entity Tag = Undead

DO

Corruption +2

OPEN >

03

TRIGGER

ON QUEST COMPLETED

1 ERROR

IF

If Quest = dark_ritual

DO

Corruption +10

OPEN >

RULE ORDER DOES NOT CHANGE EXECUTION ORDER UNLESS THE RUNTIME DECLARES IT.

INTENT

Explain automatic behavior in plain language before creators enter the graph.

KEY STATES

Enabled/disabled, invalid, reordered, no-condition, duplicate, and empty-list states.

BUILD NOTES

Generate summaries from graph nodes; never store separate summary text that can drift from the rule.

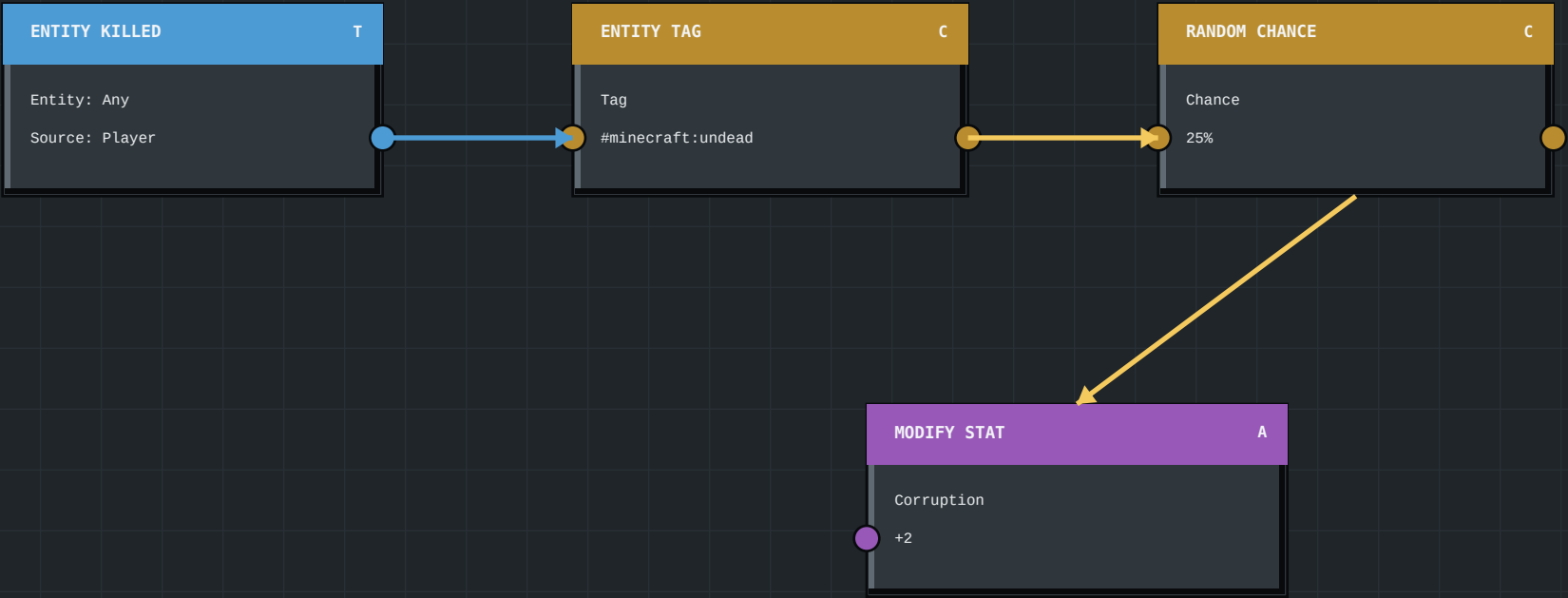
RULE 02 / UNDEAD KILL

UNSAVED

UNDO

VALIDATE

DONE



INTENT

Build complex behavior while keeping flow direction and node categories immediately legible.

KEY STATES

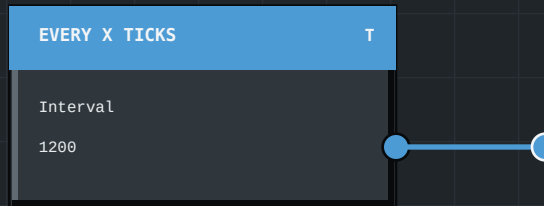
Pan/zoom, select, connect, disconnect, multi-select, minimap, undo, invalid connection, dirty graph.

BUILD NOTES

Ports accept typed connections. Category is conveyed by header color, one-letter badge, and icon - never color alone.

82% FIT





ADD COMPATIBLE NODE

FLOW OUTPUT

stat

ALL

CONDITION

ACTION

FLOW

VALUE

C

STAT CONDITION

Compare a stat value

A

MODIFY STAT

Add, subtract, or set

T

STAT CHANGED

Start when value changes

T

STAGE ENTERED

Start on stage transition

T

STAGE EXITED

Start on stage transition

ENTER PLACE / ESC CANCEL / UP-DOWN SELECT

INTENT

Turn a dragged connection or keyboard search into a short, compatible result list.

KEY STATES

Port-created picker, empty-canvas picker, recent nodes, category filter, no results, raw ID search.

BUILD NOTES

When opened from a port, compatibility is pre-filtered and the connection completes after node placement.

RULE 03 / DARK RITUAL

3 ISSUES

ISSUES 3

VALIDATION

ERROR

Unknown node type

myaddon:soul_check

ERROR

Required stat missing

Modify Stat

WARNING

Action is disconnected

Modify Stat

FOCUS SELECTED

COPY RAW JSON

INTENT

Keep broken graphs visible and repairable when dependencies or referenced content disappear.

KEY STATES

Missing field, unknown node type, disconnected trigger, invalid connection, warning-only issue.

BUILD NOTES

Never delete unknown nodes on load. Preserve their raw payload and provide copy JSON / replace-node recovery actions.

QUEST COMPLETED

T

Quest
dark_ritual

UNKNOWN NODE

U

myaddon:soul_check
Payload preserved

ADDON NOT LOADED

MODIFY STAT

A

Stat: <missing>
+10

REQUIRED STAT MISSING

 CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

● NO ERRORS

DISPLAY

How players see this stat.

☒ Show on HUD

☒ Show numeric value

DISPLAY TYPE

Bar

VISIBILITY

When non-default

LIVE PREVIEW

 CORRUPTION

60 / 100

POSSESSED

PREVIEW VALUE

60

0

100

ACTIVE STAGE

Possessed

INTENT

Configure player-facing output while previewing stage colors at arbitrary values.

KEY STATES

Bar, number, percent, icon stacks, hidden, visibility rules, no-HUD, stage transition.

BUILD NOTES

Preview value is sandboxed and never mutates player data. Preview canvas should match actual HUD scaling and safe-area constraints.

CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

NO ERRORS

ADVANCED

Source file, references, and validation.

FILE

data/mypack/rpg/stats/corruption.json

OPEN JSON

RELOAD DISK

Last loaded 12s ago

REFERENCES

14 objects reference this stat.

VIEW REFERENCES

JSON PREVIEW

```
{
  "id": "mypack:corruption",
  "value": { "default": 0, "min": 0, "max": 100 },
  "stages": [ ... ]
}
```

COPY

EDIT RAW

VALID

INTENT

Expose source-of-truth details and recovery tools without crowding normal creation.

KEY STATES

Read-only preview, raw edit, syntax error, reload conflict, copied JSON, file missing.

BUILD NOTES

Default to read-only. Raw edit requires parse + schema validation before applying and retains last valid buffer for recovery.

CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

NO ERRORS

VALIDATION

2 ERRORS

Fix errors before saving. Warnings may be intentional.

ERROR

STAGES

Stage 'corrupted' overlaps 'tainted'

Range 20 - 24

GO >

ERROR

RULES

Required stat is missing

Rule 03 / Modify Stat

GO >

WARNING

STAGES

No stage covers values 50 - 59

Gap in timeline

GO >

WARNING

GENERAL

Texture cannot be verified on server

corruption.png

GO >

CLOSE

SAVE

INTENT

Centralize save blockers and warnings with one-click navigation to the responsible control.

KEY STATES

Error vs warning, grouped by page, save blocked, save-with-warnings, stale dependency, all-clear.

BUILD NOTES

Errors block save. Warnings require acknowledgment but do not imply invalid JSON. Each issue stores a stable editor focus target.

 CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

 NO ERRORS

TEST STAT

DEV TOOL

This changes live player data and may trigger effects or rules.

CURRENT PLAYER

myyyst

ONLINE

CURRENT VALUE

37

CORRUPTED

QUICK ADJUST

-10

-1

+1

+10

SET TO MIN

SET TO MAX

RESET

LAST EVENT

Stage entered: Corrupted

INTENT

Change real test-player data deliberately and observe runtime reactions immediately.

KEY STATES

No player selected, permission denied, offline player, clamp reached, effects firing, reset confirmation.

BUILD NOTES

Use a distinct developer-tool treatment. Every mutation is explicit, logged, and restricted by server permissions.

CORRUPTION

UNSAVED

UNDO

SAVE

GENERAL

VALUE

STAGES

RULES

DISPLAY

ADVANCED

MODE

ADVANCED

CTRL+Z UNDO

CTRL+S SAVE

VALIDATION

NO ERRORS

STAGE SIMULATION

SAFE PREVIEW

PREVIEW VALUE

67

0

100



ACTIVE STAGE

POSSESSED

50 - 79

ACTIVE EFFECTS



ATTACK DAMAGE

+20%



MAX HEALTH

-4



WITCH PARTICLES

20 ticks



Preview enter/exit events



Render visual effects

INTENT

Preview stage transitions and active effects safely without touching any player.

KEY STATES

Threshold crossing, gap, overlapping stages, decimal scrub, event preview, particle/screen effects toggles.

BUILD NOTES

Evaluate against an isolated preview context. Label unavailable runtime-only effects instead of pretending they rendered.

REFERENCES TO mypack:corruption

14 TOTAL

CLOSE

Filter references...

DIALOGUE

4

Dark Priest

mypack:dark_priest.dialogue

OPEN >

Cursed Guard

mypack:cursed_guard.dialogue

OPEN >

QUESTS

5

Corruption Intro

mypack:corruption_intro

OPEN >

Embrace Darkness

mypack:embrace_darkness

OPEN >

ABILITIES

2

Blood Cast

mypack:blood_cast

OPEN >

ZONES

3

Cursed Forest

mypack:cursed_forest

OPEN >

INTENT

Answer where a stat is used before creators rename, migrate, or delete it.

KEY STATES

Grouped by object type, filter, open target, stale reference, migration preview, zero references.

BUILD NOTES

Reference indexing should be derived from loaded definitions. Show both display names and resource IDs for reliable navigation.

IMPORT STAT JSON

Paste a definition or choose a file from the current project.

CHOOSE FILE

corruption.json

JSON

```
{
  "id": "mypack:corruption",
  "value": {
    "default": 0, "min": 0, "max": 100
  },
  "myaddon:soul_decay": 0.4
}
```

☒ Keep unknown fields

CANCEL

IMPORT PREVIEW

ID

mypack:corruption

STATUS

VALID

UNKNOWN FIELDS

1 PRESERVED

IMPORT STAT

INTENT

Bring existing definitions into the same editor without hiding addon fields or parse problems.

KEY STATES

File/paste source, parse error, duplicate ID, unknown fields preserved, replace/create-copy choice, import preview.

BUILD NOTES

Treat import as validation plus save. Unknown fields receive a preservation badge and remain untouched unless a field owner is installed.

 UNSAVED CHANGES

Corruption has changes
that are not saved.

DISCARD

CANCEL

SAVE

 CHANGE RESOURCE ID

14 objects reference
mypack:corruption.

Migrate references?

☒ Update 14 references

A backup is created before
migration.

CANCEL

CHANGE ONLY

MIGRATE

 DELETE CORRUPTION?

Referenced by 14 objects:
4 dialogue / 5 quests
2 abilities / 3 zones

THIS MAY BREAK:
dialogue conditions
quests and abilities

VIEW REFS

CANCEL

DELETE

INTENT

Make unsaved, referential, and destructive decisions explicit without modal fatigue.

KEY STATES

Unsaved close, ID rename, referenced delete, external reload conflict, migration available/unavailable.

BUILD NOTES

Default focus stays on the safe action. Dangerous buttons are red, separated, and never triggered by Enter unless explicitly focused.

EDITOR COMPLEXITY

BASIC

DEFAULT

Fast setup for common stats. Advanced data is preserved but hidden.

● GENERAL

● VALUE

● STAGES

● DISPLAY

ADVANCED

Full logic, source, references, debugging, and simulation tools.

● RULES + EVENTS

● COMPUTED VALUES

● RAW JSON

● DEBUG TOOLS

ESSENTIAL SHORTCUTS

CTRL+S

Save

CTRL+Z

Undo

CTRL+Y

Redo

SPACE

Pan graph

F

Focus selection

TAB

Add node

INTENT

Reveal complexity progressively while preserving one stat definition and one save path.

KEY STATES

Mode switch, hidden advanced page with active errors, keyboard help, first-run education, per-user preference.

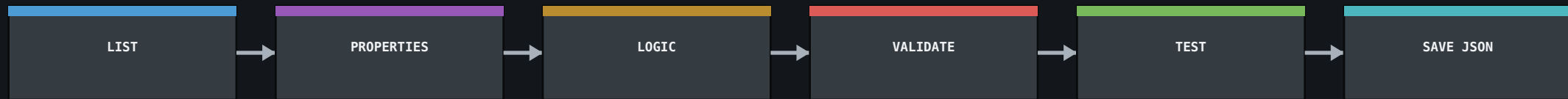
BUILD NOTES

Mode affects visibility only. Never discard advanced fields when saving from Basic mode, and always surface blocking hidden errors.

COMPLETE WORKFLOW

CREATE SIMPLY. REVEAL POWER WHEN NEEDED.

Every screen in this system serves the same promise: a creator can build Mana in seconds and still build a world-changing Corruption mechanic without leaving the game.



SOURCE OF TRUTH

`data/mypack/rpg/stats/corruption.json`